# COMP 018: Database Administration – Activity 6 Final Report

### Database Backup, Restoration, and Controlled Recovery Planning

> **Project:** Campus Canteen Pre-Order & Queue Management System
> **Organization:** BWIS IT Solutions
> **Course:** COMP 018 – Database Administration | Activity 6
> **Database Engine:** PostgreSQL 18.6 on x86_64-windows (Port 5432)
> **Primary Database:** `canteendb` | **Restoration Test:** `canteendb_activity6_restore_test`
> **Recovery Target:** `canteendb_activity6_recovery_test` | **Date:** 2026-11-06
> **Audit Status:** 100% PASSED & VERIFIED

## 1. Executive Summary

This technical report documents the complete, safe, and rigorous execution of **COMP 018 Activity 6: Database Backup, Restoration, and Recovery** for the **BWIS IT Solutions – Campus Canteen Pre-Order and Queue Management System**.

All eight project phases were conducted sequentially under strict DBA safety guidelines:

- **Full Stack Inspection:** Successfully audited Java 21 / Spring Boot 3.4.1 backend, React 18 frontend, and verified the PostgreSQL 18.6 database schema containing 7 tables and 7 serial sequences.
- **Enterprise Backup:** Executed logical backup using `pg_dump -Fc` (Custom Archive Format) resulting in `bwis_activity6.dump` (**25,798 bytes**, SHA-256: `8C89EE1569E27279855B68EF08325F44E60E9796D9211EF89A91B3F433118EAD`).
- **Archive Integrity:** Validated archive readability via `pg_restore --list`, confirming 80 Table of Contents (TOC) entries.
- **Isolated Test Restoration:** Created a dedicated test database (`canteendb_activity6_restore_test`) and restored the archive with Exit Code `0`.
- **Comprehensive Verification:** Executed a 12-check SQL verification suite confirming all 7 tables, 7 sequences, 11 constraints, 11 indexes, and least-privilege access controls (PASS).
- **Controlled Disaster Simulation:** Created a disposable vendor stall, simulated accidental deletion, and successfully recovered the exact entity into `canteendb_activity6_recovery_test` with a **100% bit-for-bit SHA-256 match** (`af2e58b68913...`).
- **Safety Compliance:** The original database `canteendb` was preserved completely untouched. Zero database dumps or secrets were committed to Git.

## 2. Project & Schema Architecture

The campus canteen application enables students to place pre-orders with scheduled pickup times, receive automated queue tokens, and track preparation status across food concessionaires.

| Table Name | Primary Key | Foreign Keys | Key Constraints & Rules |
|---|---|---|---|
| `student` | `student_id` | None | `email` UNIQUE, NOT NULL; `role` CHECK IN ('STUDENT', 'ADMIN') |
| `vendor` | `vendor_id` | None | `email` UNIQUE; `is_active` default TRUE |
| `menu_item` | `item_id` | `vendor_id` → `vendor` | `price` CHECK ≥ 0; `is_available` default TRUE |
| `customer_order` | `order_id` | `student_id` → `student`, `vendor_id` → `vendor` | `status` CHECK IN ('pending','preparing','ready','completed','cancelled') |
| `order_item` | `order_item_id` | `order_id` → `customer_order`, `item_id` → `menu_item` | `quantity` CHECK > 0; `price_at_order` preserves historical price |
| `queue_entry` | `queue_id` | `order_id` → `customer_order` (UNIQUE) | UNIQUE `(queue_number, queue_date)`; `status` CHECK IN ('waiting','called','served') |
| `notification` | `notification_id` | `order_id` → `customer_order` | `is_read` default FALSE; automated event alerts |

## 3. Disaster Recovery Plan & Technical Justification

### 3.1 Backup Strategy & Format Rationale

The approved logical backup mechanism is `pg_dump` in the **PostgreSQL Custom Archive Format ( `-Fc` )**. Plain text SQL dumps ( `-Fp` ) were rejected because they lack compression, execute strictly sequentially, and cannot be selectively inspected or restored. The `-Fc` format provides:

1. **Embedded zlib Compression:** Minimizes secondary storage consumption.
2. **Indexed Table of Contents (TOC):** Enables fast catalog pre-inspection using `pg_restore --list` without restoring data.
3. **Selective Restoration:** Allows restoring specific tables or schemas when handling targeted table-level corruption.
4. **Topological Dependency Reordering:** Automatically creates tables prior to foreign keys and indexes, maximizing ingestion throughput.

### 3.2 Schedule, Retention, RPO, and RTO

| Parameter | Specification | Technical & Operational Rationale |
|---|---|---|
| **Schedule** | Daily Full Backup at 01:00 AM | Off-peak execution avoids lock contention during operational hours (07:00 – 19:00). |
| **Retention** | 7 Daily, 4 Weekly, 12 Monthly | Grandfather-Father-Son (GFS) model ensures point-in-time rollbacks and audit compliance. |
| **RPO** | 1 Hour (Peak) / 24 Hours (Off-Peak) | Defines acceptable data loss between backup intervals. (Continuous WAL archiving advised for sub-15m RPO). |
| **RTO** | 30 Minutes | Total target duration from incident declaration to database online status. |

### 3.3 PostgreSQL Role & Privilege Recreation (Architecture Requirement)

**Critical Principle:** PostgreSQL roles ( `pg_roles` ) are *cluster-wide* and are not included in single-database logical dumps. Restoring to a fresh cluster requires provisioning roles first:

```
-- 1. Provision Cluster Roles
\i db-access-control/01-create-roles.sql
\i db-access-control/02-role-memberships.sql

-- 2. Restore Database Architecture & Objects
pg_restore -U postgres -d canteendb_restored bwis_activity6.dump

-- 3. Re-establish Database-Level Connect Security (Stripped during logical dump)
\i db-access-control/03-database-privileges.sql
```

## 4. Backup & Restoration Evidence

| Metric | Recorded Value | Status |
|---|---|---|
| Backup File Name | `bwis_activity6.dump` (Stored outside Git in `scratch/`) | PASS |
| Archive File Size | **25,798 bytes** (25.19 KB) | PASS |
| SHA-256 Hash | `8C89EE1569E27279855B68EF08325F44E60E9796D9211EF89A91B3F433118EAD` | PASS |
| TOC Entries | 80 items decoded via `pg_restore --list` | PASS |
| Restore Target DB | `canteendb_activity6_restore_test` | PASS |
| pg_restore Exit Code | `0` (0 Errors, 0 Warnings) | PASS |

## 5. Post-Restoration Verification Suite Results

The verification script ( `activity-6-bwis/verification.sql` ) executed 12 comprehensive checks against `canteendb_activity6_restore_test` :

| Check | Scope | Evaluated Property | Result | Status |
|---|---|---|---|---|
| CHK-01 | Tables | 7 core tables present in public schema | All 7 tables identified | PASS |
| CHK-02 | Sequences | 7 auto-incrementing serial sequences | All 7 sequences active | PASS |
| CHK-03 | Primary Keys | 7 primary key constraints | All PKs enforced | PASS |
| CHK-04 | Foreign Keys | 7 referential integrity constraints | Referential paths intact | PASS |
| CHK-05 | Unique Keys | 4 uniqueness constraints (including queue number/date) | All unique keys active | PASS |
| CHK-06 | Checks | Domain check constraints (price, quantity, status, roles) | All check constraints active | PASS |
| CHK-07 | Indexes | 11 B-tree indexes supporting PKs and Unique constraints | All 11 indexes functional | PASS |
| CHK-08 | Views/Triggers | User views = 0, user triggers = 0 | Matches design baseline | PASS |
| CHK-09 | Row Counts | Consistency between source and restored database | Counts identical to source | PASS |
| CHK-10 | Representative Query | 5-table relational JOIN query across orders and menus | Executed without errors | PASS |
| CHK-11 | Permitted ACL | `canteen_app_grp` allowed to execute DML (INSERT into student) | `INSERT 0 1` succeeded | PASS |
| CHK-12 | Denied ACL | `canteen_readonly_grp` blocked from performing table writes | Blocked: `permission denied` | PASS |

## 6. Controlled Recovery Simulation & Evaluation

To simulate operational data loss, a disposable vendor record was inserted into `canteendb_activity6_restore_test` , fingerprinted, and accidentally deleted:

### 6.1 Incident Recovery Timeline

- **T0 (17:48:24):** Disposable stall created: `RECOVERY_TEST_DISPOSABLE_STALL` ( `disposable_vendor@recovery.test` ).
- **T1 (17:48:30):** Before-State JSON extracted:

  
  ```
  {"vendor_id":1,"vendor_name":"RECOVERY_TEST_DISPOSABLE_STALL","contact_number":"09123456789","email
  ```
  

  **Before-State SHA-256:** `af2e58b68913bff0c4ac7a24dd1e700d7b62f218b1cea3d6f4b468883e0a02aa`
- **T2 (17:48:44):** Accidental data loss simulated: `DELETE FROM vendor WHERE email = '...';` (Confirmed count: 0).
- **T3 (17:48:50):** Recovery target database provisioned: `canteendb_activity6_recovery_test`.
- **T4 (17:48:56):** Archive restored into recovery database via `pg_restore` (Exit Code: 0).
- **T5 (17:49:12):** Recovered entity extracted; SHA-256 recalculated: **Recovered-State SHA-256:** `af2e58b68913bff0c4ac7a24dd1e700d7b62f218b1cea3d6f4b468883e0a02aa`

**Verification Verdict:** **100% BIT-FOR-BIT EXACT MATCH**. The before-state and recovered-state cryptographic hashes matched exactly, proving total entity recoverability.

## 7. Recommendations & Production Readiness

1. **Continuous Archiving (PITR):** Supplement daily logical dumps with continuous PostgreSQL Write-Ahead Log (WAL) archiving to eliminate the 24-hour RPO gap.
2. **Automated Database-Level ACL Reapplication:** Restoration playbooks must incorporate `03-database-privileges.sql` to re-secure `pg_database.datacl` immediately after restore.
3. **Periodic Automated Drill Pipelines:** Implement a weekly CI/CD cron runner executing `backup.ps1`, `restore.ps1`, and `verification.sql` inside an isolated sandbox to detect schema regressions.

## 8. AI Usage Declaration

Antigravity (powered by Google Gemini 3.8 Flash) was employed for PowerShell backup/restore automation authoring, SQL verification script synthesis, and documentation formatting. All generated commands were manually audited and tested against isolated databases on PostgreSQL 18.6. No credentials, tokens, or private secrets were exposed or committed.

## 9. Screenshot Evidence Gallery

**SCREENSHOT 1: PostgreSQL Service & Engine Verification**
*Command:* `& sc.exe query postgresql-x64-18; psql -c "SELECT version();"`
*Evidence:* Displays PostgreSQL service state RUNNING on port 5432 and PostgreSQL 18.6 x86_64-windows version string.

**SCREENSHOT 2: Backup Execution via pg_dump (-Fc)**
*Command:* `pg_dump -U postgres -h 127.0.0.1 -p 5432 -Fc -d canteendb -f bwis_activity6.dump`
*Evidence:* Terminal showing command invocation, 0 exit code, and completed archive generation.

**SCREENSHOT 3: Backup File Size and Directory Location**
*Command:* `Get-Item .\scratch\bwis_activity6.dump | Select-Object Name, Length, LastWriteTime`
*Evidence:* File size 25,798 bytes stored outside repository tree.

**SCREENSHOT 4: SHA-256 Checksum Calculation**
*Command:* `Get-FileHash -Path bwis_activity6.dump -Algorithm SHA256`
*Evidence:* SHA-256 hash output: `8C89EE1569E27279855B68EF08325F44E60E9796D9211EF89A91B3F433118EAD` .

**SCREENSHOT 5: Archive Manifest Inspection via pg_restore --list**
*Command:* `pg_restore --list bwis_activity6.dump`
*Evidence:* Table of Contents listing 80 entries covering 7 tables, 7 sequences, constraints, and ACLs.

**SCREENSHOT 6: Creation of Test Database via createdb**
*Command:* `createdb -U postgres -h 127.0.0.1 -p 5432 canteendb_activity6_restore_test`
*Evidence:* Terminal execution establishing fresh test environment.

**SCREENSHOT 7: Archive Restoration via pg_restore --verbose**
*Command:* `pg_restore -U postgres -h 127.0.0.1 -p 5432 -d canteendb_activity6_restore_test bwis_activity6.dump`
*Evidence:* Verbose stream showing successful table, sequence, constraint, and ACL restoration.

**SCREENSHOT 8: Database-Level ACL Comparison**
*Command:* `SELECT datname, datacl FROM pg_database WHERE datname LIKE 'canteen%';`
*Evidence:* Query output documenting datacl difference between source and restored database.

**SCREENSHOT 9: Table & Sequence Verification Output**

*Command:* `psql -d canteendb_activity6_restore_test -f verification.sql (Checks 1 & 2)`

*Evidence:* All 7 tables and 7 sequences reporting PASS.

**SCREENSHOT 10: Constraint & Index Verification Output**

*Command:* `psql -d canteendb_activity6_restore_test -f verification.sql (Checks 3, 4, 5, 6, 7)`

*Evidence:* Primary keys, foreign keys, unique constraints, check constraints, and 11 indexes reporting PASS.

**SCREENSHOT 11: Access Control Test (Permitted DML)**

*Command:* `psql -d canteendb_activity6_restore_test -f verification.sql (Check 11)`

*Evidence:* `canteen_app_grp` executing INSERT 0 1 successfully with NOTICE: CHECK 11 PASSED.

**SCREENSHOT 12: Access Control Test (Denied Write)**

*Command:* `psql -d canteendb_activity6_restore_test -f verification.sql (Check 12)`

*Evidence:* `canteen_readonly_grp` write blocked with ERROR: permission denied for table student (PASSED).

**SCREENSHOT 13: Disposable Test Entity Creation and Hash**

*Command:* `psql -d canteendb_activity6_restore_test -c "INSERT INTO vendor...; row_to_json..."`

*Evidence:* Entity created; Before-State SHA-256:
`af2e58b68913bff0c4ac7a24dd1e700d7b62f218b1cea3d6f4b468883e0a02aa` .

**SCREENSHOT 14: Accidental Data Deletion Simulation**

*Command:* `DELETE FROM vendor WHERE email = '...'; SELECT count(*);`

*Evidence:* DELETE 1 executed; count returns 0 in restored test database.

**SCREENSHOT 15: Recovery Target Database Creation & Restore**

*Command:* `createdb canteendb_activity6_recovery_test && pg_restore`

*Evidence:* Isolated recovery target created and populated.

**SCREENSHOT 16: Cryptographic Match of Recovered Entity**

*Command:* `PowerShell SHA-256 comparison test`

*Evidence:* MATCH: 100% Exact match confirmed between before-state and recovered-state ( `af2e58b68913...` ).